Python CGPA Calculator

Overview

This project is a to-use and beginner-level Python application that assists students in computing their CGPA on a 10-point scale. You simply input the marks and credits for every subject. The program takes care of the rest.

It checks the inputs transforms your marks into grade points and computes the CGPA precisely. Whether you’re a student monitoring your performance or an institution seeking a fast method to calculate CGPA this tool accomplishes the task easily.

Features

• Converts percentage marks into grade points

(10 for 90+, 9 for 80–89, 8 for 70–79, and so on)

• Works for any number of subjects or courses

• Ensures marks are within 0–100 and credits are positive

• Handles invalid or non-numeric inputs by skipping those subjects safely

• Displays the CGPA in a neat and straightforward layout

• Notifies the user when no valid subjects have been entered

Utilized Technologies

• Python 3 (tested on Python 3.6 and above)

• Does not require libraries. Relies solely, on built-in Python capabilities

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How to Install & Run

1. Ensure Python 3 is set up on your computer.

2. Save the program file as cgpa_calculator.py.

3. Open a terminal and navigate (cd) to the folder containing the file.

4. Execute the program with: CGPA Calculator.py

5. Follow the on-screen instructions to enter:

• Number of subjects

• Marks (0–100)

• Credit for each subject

Testing Guide

Try the following cases to thoroughly test the program:

Normal Cases

• Valid marks (0–100)

• Positive credits

• Verify if the CGPA has been computed accurately

Invalid Input Cases

• Marks less than 0 or greater than 100

• Credits that are zero or negative

• Non-numeric inputs for marks or credits

Edge Case

• Enter zero valid subjects

• The program must indicate that calculating CGPA is not possible

Optional Screenshots

• A successful run showing the CGPA output

• An example showing an error message for invalid marks

• A run where a subject is skipped due to invalid input